

SIGNALBRIDGE

Turning Emergency Warnings into Personalized, Accessible Actions

Don't just receive the warning. Understand your risk. Know your next safe action.

1. PROBLEM STATEMENT

Emergency warnings communicate that a hazard exists, but people may still not know:

- Am I affected?
- What should I do now?
- What should I avoid?
- Which information can I trust?

People with disabilities, low digital literacy, and regional-language preferences can face additional barriers.

Core gap: Warning → Understanding → Action

2. PROPOSED SOLUTION

SignalBridge is an AI-powered alert-to-action system that converts verified emergency warnings into personalized, source-backed action plans.

It answers:

- **WHAT?** What is happening?
- **AM I AFFECTED?** Is my area within the warning zone?
- **WHAT NOW?** What should I do immediately?
- **WHAT NEXT?** What should I do in the next hour?
- **WHAT IS VERIFIED?** Which information is official or uncertain?

Outputs include **English, Tamil, voice, maps, and accessible action cards.**

3. RESEARCH & EXISTING GAP

India already has systems such as **SACHET** and **IMD** for authoritative disaster warnings.

SignalBridge **does not replace these systems.** It adds an AI interpretation layer after a trusted warning is received.

Existing flow:

Warning → Person

SignalBridge flow:

Warning → Evidence → Relevance → Personalization → Action

The concept is informed by people-centred early-warning principles from **WMO, UNDRR and WHO.**

4. KEY INNOVATION

Alert-to-Action Compiler

Warning → Evidence → Relevance → Personalization → Action

Personal Relevance: Determines whether the user's selected area is affected.

Evidence-Linked AI: Connects generated actions to trusted guidance.

Uncertainty Layer: Separates official, community-reported and conflicting information.

Accessibility Layer: Provides simplified language, Tamil, voice and accessible visual instructions.

Citizen Intelligence: Clusters duplicate reports while keeping unverified reports clearly labelled.

5. HOW IT WORKS

Emergency Warning + Citizen Reports

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Source & Evidence Verification

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Geospatial Relevance

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Duplicate / Conflict Detection

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Trusted Guidance + RAG

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AI Personalization

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Personalized Action

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Text | Tamil | Voice | Map

6. MVP & LIVE DEMO

MVP

- **Hazard:** Floods
- **Region:** One selected city/district
- **Languages:** English + Tamil
- **Input:** Controlled/simulated emergency warning

Demo

1. Load a flood warning.
2. Identify whether the selected area is affected.
3. Generate **“Do Now”** and **“Next 1 Hour”** actions.
4. Display the affected area on a map.
5. Cluster duplicate citizen reports.
6. Label unverified information.

7. Provide Tamil voice output.
8. Show the evidence behind the generated action.

7. TECHNOLOGY

Frontend: React, PWA, Leaflet, accessibility-first UI

Backend: Python, FastAPI, SQLite/PostgreSQL

AI: RAG, NLP classification, semantic clustering, contradiction detection, multilingual processing, Text-to-Speech

Geospatial: GeoJSON, GeoPandas, Shapely, Leaflet

8. IMPACT & SCALABILITY

Prototype Metrics

- Warning comprehension
- Action-selection time
- Geographic relevance
- Source attribution accuracy
- Duplicate-report detection
- Uncertainty classification
- Accessibility coverage

Future Expansion

Floods → Cyclones → Heatwaves → Fires → Landslides

English + Tamil → Multiple Indian languages

Individuals → Communities → NGOs → Authorized emergency ecosystems

FINAL VALUE PROPOSITION

SignalBridge doesn't create another warning. It creates the bridge between receiving a warning and knowing your next safe action.